

Nicolas Luthi

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SUMMARY

Software Engineer with hands-on experience architecting secure, reliable web and mobile systems, delivering end-to-end solutions using C++, Java, Python, and SQL. Strong background in teamwork, technical communication, problem solving, and systems development; skilled in RESTful APIs, authentication, multithreading, and software architecture.

EXPERIENCE

Code Instructor

iCode Scottsdale

Feb 2026 – Present

Scottsdale, AZ

- Deliver structured technical instruction across C++, Python, C#, and Java, emphasizing repeatable outcomes via milestone-driven, iterative execution and clear evaluation criteria.
- Lead C++ Arduino robotics builds and Python DroneKit projects, guiding end-to-end hardware–software integration, systematic troubleshooting, and validated fixes against functional objectives.
- Track progress against defined requirements/objectives, adjust plans to close gaps, and reduce recurring issues through consistent debugging and process improvement.
- Communicate technical status and results to parents and administrators, demonstrating strong customer-facing collaboration, presentation, and clarity across technical/non-technical stakeholders.

Software Development Intern

Healthee, Inc.

June 2023 – Dec 2023

Scottsdale, AZ

- Improved response relevance and consistency by integrating OpenAI APIs with 10,000+ curated training data points; iterated implementation based on user feedback and measured outcomes.
- Implemented secure authentication and data management using Firebase, supporting controlled access, reliable sessions, and consistent user experiences.
- Built and maintained a cross-platform Flutter UI enabling real-time interactions; documented technical tradeoffs and implementation decisions to improve maintainability.
- Collaborated with product and engineering to translate requirements into validated deliverables using an iterative development approach and disciplined version control practices.

PROJECTS

Music Sharing Platform | React, Django, REST Framework, PostgreSQL

Spring 2026

- Built an OAuth 2.0-integrated backend connecting Spotify and Apple Music, emphasizing secure access control, reliable API behavior, and clear integration boundaries.
- Designed a normalized PostgreSQL schema with indexing and query optimization to support traceable user relationships, engagement metrics, and weekly analytics.
- Delivered real-time dashboards and notifications via WebSockets, improving timeliness of updates and supporting validation through observable system events.

Service Oriented Web Application | C#, ASP.NET Framework, XML, JavaScript, IIS Express

Spring 2025

- Built a multi-tier application separating presentation, business logic, and data layers to support controlled changes and easier verification of system behavior.
- Implemented custom authentication and access control, securely storing encrypted credentials in XML and reducing risk through reusable security modules (C# DLL).
- Added session state management and cookies to preserve consistent authenticated behavior and reduce user-facing errors.

Desktop Medical Records Application | Java, JavaFX, MySQL, JUnit

Spring 2025

- Designed a multi-user system with role-based access to sensitive records, emphasizing correctness, controlled data access, and reliable persistence.
- Built secure messaging and database integration components supporting core workflows (records, prescriptions, appointments).

MapReduce Word Count System | Python, Flask, Multithreading

Spring 2024

- Developed a Flask web application implementing a MapReduce pipeline for word-frequency analysis on uploaded text files.
- Implemented thread-safe aggregation with locks to support concurrent map/reduce execution and correct results under parallel workloads.

TECHNICAL SKILLS

Languages: C/C++, Python, C#, Java, Dart, JavaScript, XML

Software Concepts & Methodologies: Agile, Software Architecture, Digital Design, Software QA and Testing, Software Documentation

Systems & Platforms: Linux, iOS, macOS, Android, Windows, Cross-Platform Mobile and Web Development

Tools: Git, GitLab, Jira, Taiga, Claude Code, Codex, Cursor, Gemini Code

Frameworks: ASP.NET Framework, Flask, FastAPI, Flutter, LangChain, Xamarin, IIS Express

Databases & Cloud: Firebase, PostgreSQL, Google Cloud Platform

EDUCATION

Arizona State University

Bachelor of Science in Computer Science

Aug 2023 – May 2026

Tempe, AZ